# Mira Askar

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## EDUCATION

**University of Warwick** — **Coventry, United Kingdom**
*BEng Electrical and Electronic Engineering* — Sep 2025 - Present
Expected graduation: July 2028 | First-year average: 74.8% (First Class) | Engineering Mathematics: 91%

**Malvern College** — Sep 2023 - Aug 2025
*A Levels:* Mathematics (A), Physics (A), Russian (A*)

## TECHNICAL SKILLS

**Languages:** Java, SQL, C/C++, MATLAB
**Frameworks & Testing:** Spring Boot, Hibernate ORM, JUnit
**Tools & Technologies:** PostgreSQL, Docker, Git, REST APIs, JSON, Simulink
**Hardware & Embedded:** STM32, Arduino, KiCad, LTspice, CAN, UART

## WORK EXPERIENCE

**Healware** — **Hong Kong (Remote)**
*Java Backend Developer* — Aug 2026 - Sep 2026

- Built a prototype Java/Spring Boot REST API (4 business endpoints + 1 technical endpoint) that ingests medication selections and patient allergy histories as JSON and returns allergy-risk assessments.
- Implemented the risk logic: direct active-ingredient matching plus rule-based cross-reactivity checks, returning contributing factors and safer medication alternatives through structured DTOs.
- Delivered the service independently with input validation, exception handling and structured error responses (correct HTTP status codes), tested with JUnit and integration HTTP tests on synthetic patient data.

**Trans Asia Construction LLP** — **Kyzylorda, Kazakhstan**
*Java Backend Developer* — Jul 2026 - Aug 2026

- Developed a Java/Spring Boot prototype for employee record management using Hibernate ORM and PostgreSQL, containerised with Docker.
- Designed a relational database schema and REST APIs for employee CRUD operations, structured search and document-status tracking, supporting a proposed transition from Excel-based record keeping.
- Implemented server-side validation, database constraints and transactional updates to maintain consistency across employee records.

## PROJECTS

**Poker Decision Simulator** — Mar 2026
*Java, JUnit, Monte Carlo Simulation*

- Built a Java Texas Hold'em simulator with complete betting rounds, hand evaluation and three rule-based opponents with distinct playing styles.
- Implemented Monte Carlo hand-equity estimation using 5,000 simulations per decision to support decisions under incomplete information.
- Developed an explainable decision engine using pot odds and estimated call EV, with rule-based raise recommendations and JUnit tests covering hand rankings, betting rules and pot distribution.

**Self-Balancing Robot** — Jun 2026
*University Engineering Project | MATLAB, Simulink, Arduino Mega 2560, PID Control, LaTeX*

- Tuned the PID balance controller on live hardware via Simulink Monitor & Tune, using binary search over gains: cut derivative gain (0.5 → 0.15) to stop overcorrection, raised integral gain to reject disturbances and added ±255 motor saturation.
- Sustained balance for 15+ minutes on carpet, a wooden desk and the lab floor, 30× the 30-second target.
- Co-authored the project documentation in LaTeX (Overleaf), tracing the V-Model from requirements to validation; project awarded First Class.

**Warwick Hyperloop** — Oct 2025 - Present
*Electronics Hardware Engineer, Battery Subteam | KiCad, LTspice, STM32, CAN, UART*

- Designed a battery-temperature monitoring PCB in KiCad (schematic and layout) around an STM32 microcontroller, reading 8 temperature sensors through multiplexers.
- Simulated the design in LTspice before prototyping, validating voltage dividers, multiplexer switching and diode/transistor stages.
- Integrated CAN and UART interfaces into the PCB design to connect the board with the BMS and the Control Systems subteam's Arduino Mega.

## ACHIEVEMENTS & INTERESTS

**Awards:** British Physics Olympiad (BPhO) - Gold; Gold CREST Award (fuel-level sensor system for QinetiQ).
**Scholarships:** University of Warwick Global Excellence Scholarship; Malvern College Honorary Scholarship.
**Sponsorship:** Secured approximately £1,000 from KT Lab Cloud and Chevron at the Enactus National Expo Kazakhstan.
**Interests:** Piano (completed music school with a final grade 5.0/5.0); choir and orchestra; Warwick volleyball; robotics.